

# Anupam Pokharel

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## EDUCATION

### Carnegie Mellon

Pittsburgh, PA | Graduated Dec. 2021

#### B.S. IN STATISTICS & MACHINE LEARNING, MINOR IN COMPUTER SCIENCE

Dean's List with High Honors (F '20, S '21); 3.59 GPA (last two years)

**Coursework:** Imperative Computation, Functional Programming, Parallel & Sequential Data Structures & Algorithms, Intro. to Computer Security, Computer Vision, Modern Regression, Advanced Data Analysis, Intro. to Machine Learning, Intro. to Deep Learning, Computer Networks, Parallel Computer Architecture & Programming

## WORK EXPERIENCE

### IBM | STAFF SOFTWARE ENGINEER, FULL-STACK

Austin, TX | Aug. 2022 -

Cloud

- Working on a team tasked with development, DevOps, and release of the IBM Cloud CLI and in-house tooling that instantly produces or updates 85%+ of source code for multi-language SDKs and CLI plug-in binaries for nearly all PaaS offerings (thereby greatly reducing the average timeframe for readying new submissions to the catalog of CLI plug-ins)
  - implemented concurrency handling, allowing CLIs to execute in parallel without corrupting client-side data
  - added support for auxiliary sessions that can temporarily alter the privileges of the currently-active session
- Helping develop several sections of IBM Cloud Projects related to clients' IaC UXs, with involvement across the stack
  - deciding data schema and writing back-end functions for APIs that execute new resource-tagging protocols
  - writing and reviewing code for the webpages deployed to [cloud.ibm.com/projects](https://cloud.ibm.com/projects) (implemented in a mix of client-side- and server-side- rendered React.js components, with a GraphQL client provided by Apollo)

### IBM | ASSOCIATE SOFTWARE ENGINEER, BACK-END

Poughkeepsie, NY | Feb. - Aug. 2022

Competitive Insights (re-branding of CPO from Summer 2021)

- Led a project on Power Systems: designing a fully automated stack for comparing shared processing adjustments by hypervisor firmware with resource-scaling by Openshift's HPA during erratic load surges created by a suite of bootstrapped applications and virtual user activity profiles that leveraged Summer 2021's work

### IBM | BACK-END DEVELOPER INTERN

Remote | May - Aug. 2021

Competitive Project Office (CPO) in Cloud and Cognitive Software Division

- Mainly worked on a project to benchmark systems orchestrated by Openshift on two architectures - IBM z/Arch. and Intel x86 - during I/O- & CPU-bound workloads (executed using a containerized Flask upload/download microservice for IBM Aspera coded from-scratch), via Javascript code generating variable emulated user activity

### CARNEGIE MELLON | UNDERGRADUATE TEACHING ASSISTANT

Pittsburgh, PA | Feb. - May 2021

Department of Math

- Led weekly recitations to review course material and co-conducted grading for the Spring 2021 offering of 21-120: Differential and Integral Calculus

## FEATURED PROJECTS

### TAGGING RESOURCES FOR IBM CLOUD PROJECTS [IBM Developer Blog](#)    **NODE.JS, REACT.JS | DEC. 2023**

A series of web-pages driving customer adoption of IBM Cloud Projects, now generally available on the IBM Cloud Console, in which users can bring into a Project resources that were provisioned outside of or before Projects

### PARALLELIZED INFERENCE ON RNNs [Project Exhibition \(GH Pages\)](#)    **C++, CUDA, BASH | DEC. 2021**

A project that explores the speedups attainable by parallelizing some parts of inference-time Recurrent Neural Network (RNN) computations, at varying extents of compromise on accuracy

## SKILLS

**Technologies and Professional Tools:** VCS (git, Gitlab/Github), Docker(hub), CI/CD (e.g. Jenkins, Travis, Tekton), Cloud resource mgmt/provisioning (mostly on IBM Cloud & AWS EC2), Setup/use of hypervisors (z/VM), Container orchestration (k8s, Openshift)

**Languages:** C++, C, Go, Java, Javascript, React.js, Ruby, Shell scripting, Intel x86 Assembly, Standard ML of NJ, Python (especially with PyTorch, Numpy, Tensorflow, and OpenCV libraries for CNNs/ML/DL), as well as more data-centric experience with R and SQL

**Conceptual and Technical Proficiency in:** Hardware virtualization, Networking protocols and security, RESTful principles and API design/use, Data structures & algorithms, Concurrency (threading, mutex, etc.), ML (high-dim. data & PCA, modeling, CV, etc.)